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Effect of Lung Volume on Voice Onset Time (VOT)

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This investigation was designed to test the hypothesis that voice onset time (VOT) varies as a function of lung volume. Recordings were made of five men as they repeated a phrase containing stressed /pi/ syllables, beginning at total lung capacity and ending at residual volume. VOT was found to be longer at high lung volumes and shorter at low lung volumes in most cases. This finding points out the need to take lung volume into account when using VOT as an index of laryngeal behavior in both healthy individuals and those with speech disorders.

KEY WORDS: voice onset time, respiratory, laryngeal

Voice onset time (VOT) is an important acoustic feature used in distinguishing voiceless from voiced stop consonants in English (Lisker & Abramson, 1964, 1970; Zlatin, 1974). VOT has been shown to have several dependencies, such as place of articulation and utterance context (Klatt, 1975; Lisker & Abramson, 1967; Port & Rotunno, 1979; Weismer, 1979). Study of such dependencies has focused on laryngeal and upper airway variables to the exclusion of potentially influential respiratory variables. Because VOT has been used as an index of the control of timing between laryngeal and upper airway articulation, particularly with respect to disordered speech production, it would seem important to determine the impact of respiratory dependencies, if any, on this measure.

One reason to believe that respiratory variables may influence VOT is that the larynx is linked mechanically to the respiratory apparatus. This linkage makes it possible for the respiratory apparatus to exert force on the larynx. For example, at high lung volumes, the diaphragm usually flattens and pulls the trachea and larynx caudally, exerting a force that tends to abduct the vocal folds (Zenker, 1964). This abductory force conceivably could delay the onset of vocal fold vibration, which then could be realized acoustically as a longer VOT. Another reason to think that there may be a respiratory effect on VOT is related to a behavioral strategy for reducing lung volume expenditure. That is, when an individual speaks at very low lung volumes, speech may be modified in an attempt to "conserve" the small amount of air remaining in the lungs. This modification might include, for example, a reduction of air expenditure during the production of plosive consonants, which in turn might be realized acoustically as a shortening of VOT. Based on such reasoning, we conducted the present investigation to test the hypothesis that VOT decreases with decreasing lung volume.

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Method

Five men, ranging in age from 20 to 24 years, were studied. They were (a) White; (b) healthy; (c) nonsmokers; (d) without surgery or injury to the speech production apparatus; (e) free of medications; (f) normal in voice, speech, language, and hearing; (g) first-language General American English speakers; and (h) without professional training or experience in public speaking, acting, or singing. The subjects were unaware of the purpose of the investigation and had no formal knowledge of speech physiology or speech acoustics.

The speech signal and surface motions of the chest wall were recorded with the subject in a standing position. The speech signal was sensed with a high-quality microphone (Shure SM10A) positioned directly in front of the subject's lips at a distance of approximately 4 cm. The microphone was secured in a head mount so that the lip-to-microphone distance was constant throughout the recording session. Speech was recorded on a high-quality tape recorder (Tascam 32).

Surface motions of the chest wall were sensed using linearized magnetometers (GMG Scientific Inc., Burlington, MA). These magnetometers included two pairs of coils, one for the rib cage and one for the abdomen. The generator coil in each pair was taped to the front of the torso, that for the rib cage at sternal midlength, and that for the abdomen just above the umbilicus. The sensor coils were taped to the back of the torso at corresponding axial levels. Output signals reflecting anteroposterior displacements of the rib cage and abdomen were recorded on two channels of an FM tape recorder (Hewlett-Packard 396A). The speech signal was recorded on another channel of the FM recorder so that speech events could be synchronized with breathing behavior.

Each subject produced a six-syllable phrase containing a stressed /pi/ in the second and fifth syllable positions. The phrase was "a peek at a peacock" except in the case of Subject 3, for whom "a peach and a pita" was used (Subject 3 was unable to produce the original phrase without articulatory errors). Each subject was instructed to inspire as much air as possible and to produce the phrase repeatedly using his normal pitch, loudness, and quality until he ran out of breath. His speech was paced using a metronome so that the stressed /pi/ syllables coincided with a beat of approximately 2/sec. Productions were judged on-line by two investigators; those judged unacceptable in voice characteristics, articulatory accuracy, or speech rate were not analyzed. Three acceptable productions were obtained from each subject (except Subject 3, from whom two were obtained).

The subject also performed isovolume maneuvers, vital capacity maneuvers, and inspiratory/expiratory maneuvers used for calibrating chest wall motion data. Isovolume maneuvers were performed by the subject closing his airway and displacing volume back and forth between the abdomen and rib cage. These maneuvers were performed at three lung volume levels: at 95% of the vital capacity, at the resting expiratory level, and at 5% of the vital capacity. Data from these maneuvers were used to determine the volume-motion relationships between the rib cage and abdomen. Vital capacity maneuvers involved having the subject inspire fully from the resting expiratory level and expire fully while coupled to a 9-liter spirometer (Warren E. Collins, Braintree, MA). For inspiratory/expiratory maneuvers, the subject inspired and expired known volumes of air from the resting expiratory level while coupled to the spirometer.

The speech data and chest wall data were analyzed to obtain measures of VOT and lung volume change, respectively. Measures of VOT were determined from hard-copy wide-band (300 Hz) spectrograms (Kay 7800 Digital Sonograph) for each of the stressed /pi/ productions. VOT was defined as the time interval from the onset of the noise burst (associated with /p/) to the onset of the appearance of the first vertical striation at F_2 (associated with /i/).

Measures of lung volume were determined from the chest wall motion data using the general procedures described by Hixon, Goldman, and Mead (1973). These procedures and their underlying theory are described in detail elsewhere (Hixon et al., 1973). Briefly, we constructed graphs displaying rib cage displacement on the vertical axis and abdominal displacement on the horizontal axis. The relative gains of the rib cage and abdominal signals were adjusted to reflect equal and opposite volume change during isovolume maneuvers performed at the resting expiratory level. Any differences in the volume-motion relationships of the rib cage and abdomen at high and low lung volumes (at 95% and 5% of the vital capacity) were minor, and any adjustments in lung volume calibration were solved graphically. These graphs were calibrated for lung volume using data from vital capacity maneuvers and inspiratory/expiratory maneuvers. The lung volume at which each /pi/ was produced was determined by listening to the synchronized audio recording and marking the chest wall data accordingly.

All data were measured by one investigator. In addition, 10% of the data were remeasured by the same investigator (at least 2 weeks after the original measurements were made) and by a second investigator. For the VOT data, a remeasure was said to agree with the original measure if the difference between them was 8 msec or less (the approximate duration between glottal pulses for the present subjects). For the lung volume data, a remeasure was said to agree with the original measure if it differed by 5 %VC or less.

Results

Reliability of measures was high. For VOT, comparison of intrajudge measures resulted in 95.1% agreement, and comparison of interjudge measures resulted in 94.3% agreement. For lung volume, both intrajudge and interjudge comparisons yielded 100% agreement.

Results are displayed graphically in Figure 1. Each graph in the figure represents data obtained for a single trial from one subject; each point within the graphs represents the VOT of one /p/ production. Note that in some cases VOT appears generally to decrease with decreasing lung volume (e.g., Subject 1, trial a), and in others there is no apparent relation of VOT to lung volume (e.g., Subject 5, trial a). To determine if there was an association between lung volume and VOT, Kendall's tau-b analyses (Kendall, 1962) were performed, the results of which are presented in Table 1. A significant

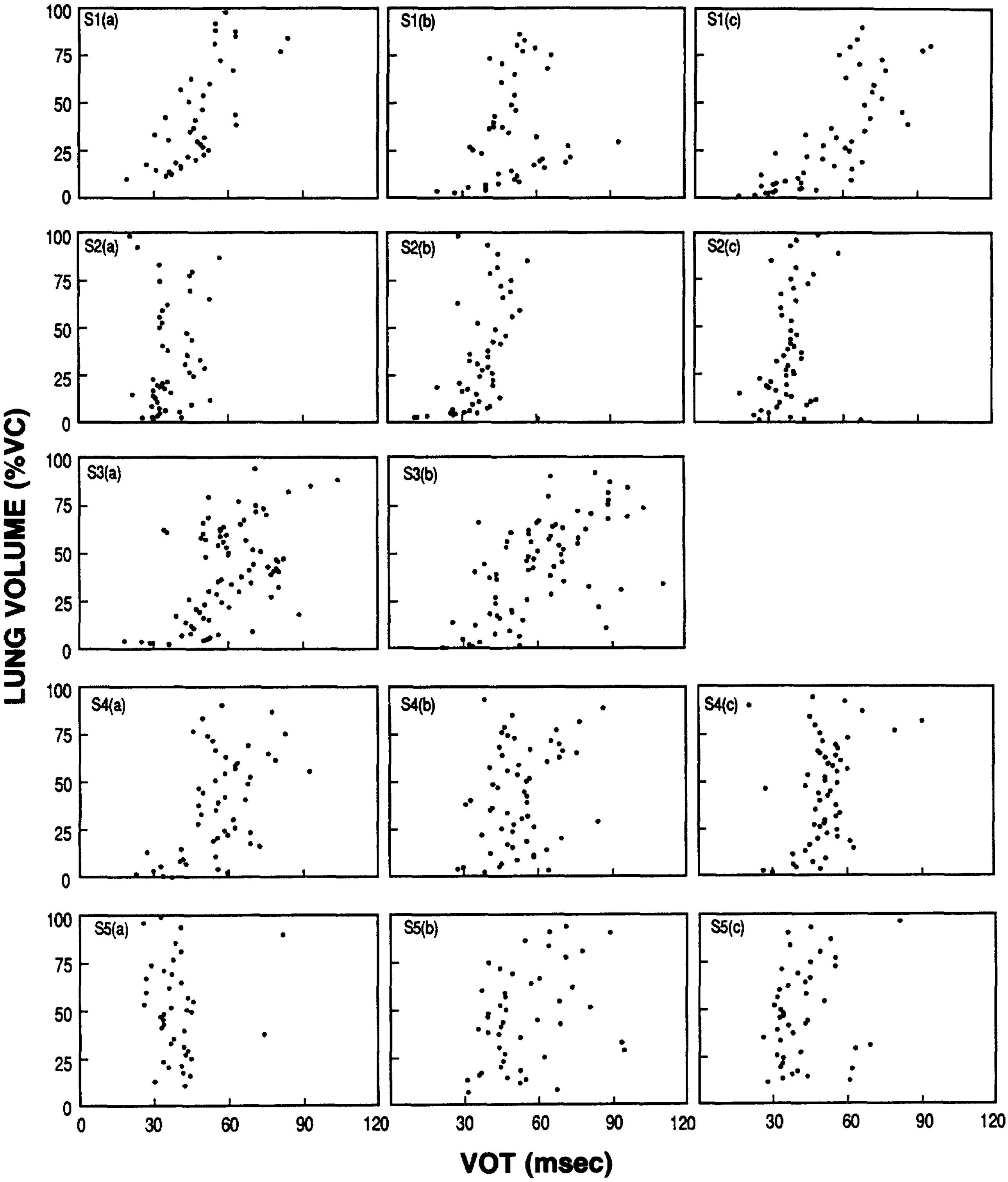

FIGURE 1. VOT values (in msec) are displayed as a function of lung volume (in percentage of vital capacity, %VC) for all trials of each subject.

concordance ($p < .05$) was found for all trials except two (trials a and c for Subject 5), indicating that there was a demonstrable association between lung volume and VOT. This association was characterized by a tendency for VOT to decrease with decreasing lung volume.

Discussion

The results of this investigation support our prediction that VOT generally is longer at high lung volumes and shorter at low lung volumes. We believe that this can best be explained by a combination of mechanical and behavioral factors, as discussed earlier. That is, the tendency for VOT to be long at high lung volumes probably can be attributed to the "tracheal tug" placed on the larynx when the diaphragm flattens, and the tendency for VOT to be short at low lung volumes may be related to a drive to conserve air.

This observation not only contributes to our understanding of normal speech physiology, but it also has potential clinical implications. There is a substantial literature documenting aberrations in VOT in subjects with disorders such as dysphasia (Blumstein, Cooper, Goodglass, Statlender, & Gottlieb, 1980; Blumstein, Cooper, Zurif, & Caramazza, 1977; Gandour & Dardarananda, 1984; Shewan, Leeper, & Booth, 1984), dyspraxia (Freeman, Sands, & Harris, 1978; Hoit-Dalgaard, Murry, & Kopp, 1983; Kent & Rosenbek, 1983), dysarthria (Caruso & Burton, 1987; Forrest, Weismer, & Turner, 1989; Kent, Netsell, & Abbs, 1979; Morris, 1989; Till & Toye, 1988; Weismer, 1984), stuttering (Healey & Gutkin, 1984; Healey & Ramig, 1989; Metz, Schiavetti, & Sacco, 1990; Prins & Hubbard, 1990), cleft palate (O'Gara & Logemann, 1982; Rolnick & Hoops, 1971), and hearing loss (Irvin & Wilson, 1973). Often abnormally long or abnormally short VOT values are interpreted to reflect impaired coordination of laryngeal and upper airway articulation. This may be true; however, it should be noted that certain disorders have been associated with the use of abnormal lung volume ranges during speech production. For example, speakers with hearing loss have been shown to use lower-than-normal lung volumes (Forner & Hixon, 1977; Whitehead, 1983). Conversely, individuals with asthma (Loudon, Lee, & Holcomb, 1988) and individuals with cervical spinal cord injury (Hoit, Banzett, Brown, & Loring, 1990) often use higher-than-normal lung volumes when they speak. In such cases, it seems apparent that lung volume events should be taken into consideration when interpreting measures of VOT.

Clearly, there is much to be learned about the influences of respiratory variables on laryngeal behavior during speech production. The present findings contribute to our understanding of those influences and point to the need to take lung volume into account when using VOT measures to address both research and clinical questions.

TABLE 1. Results of statistical analyses.

Subject	Trial	Number of tokens	Kendall's tau-b	Probability of concordance	*p*-value
1	a	40	.542	.771	.0001*
	b	43	.248	.624	.0096*
	c	47	.635	.818	.0001*
2	a	47	.257	.629	.0054*
	b	49	.482	.741	.0001*
	c	50	.316	.658	.0006*
3	a	77	.300	.650	.0001*
	b	73	.457	.729	.0001*
4	a	51	.346	.673	.0002*
	b	56	.205	.603	.0129*
	c	57	.276	.638	.0012*
5	a	39	−.030	.485	.6060
	b	43	.230	.615	.0150*
	c	43	.154	.577	.0721

$*p < .05$.

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